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MILITARY SPECIFICATION

GENERATOR, ENGINE ACCESSORY 28 VOLT DC, RATED OUTPUT 300 AMPS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a 28 volt (nominal), direct current (DC), blower-cooled generator with rated output at 300 amperes (AMPS) (see 6.1). The specification also covers repair parts for the generator.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

FF-B-171	-	Bearings, Ball, Annular (General Purpose).
QQ-P-416	-	Plating, Cadmium (Electrodeposited).
QQ-Z-325	-	Zinc Coating, Electrodeposited, Requirements for.
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-489	-	Enamel Alkyd, Gloss, (for Exterior and Interior Surfaces).
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal.

Military

MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-P-8585	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-C-11693	-	Capacitors, Feed Through, Radio-Interference Reduction, AC and DC (Hermetically Sealed in Metallic Cases), Established and Non-Established Reliability, General Specification For.

FSC-2920

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MIL-F-13927 - Fungus Resistance Test; Automotive Components.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1275 - Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles.
MS-51004 - Generator, Engine Accessory, 28 Volts, DC 300 Ampere.

DRAWINGS
Army
7717421 - Generator, Engine Accessory (M48 and M59).
10882774 - Cradle, Generator Engine Accessory Assembly.
10884044 - Adapter.
10889713 - Generator, Engine Accessory (M60A1).
10889998 - Generator, Engine Accessory (M88).
10898759 - Fan, Electrical Rotating Equipment.
10898795 - Generator, Engine Accessory.
10912456 - Brushes, Generator 300 AMPS.
10914831 - Generator, Engine Accessory (M107).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20025).

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3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2 and 6.3), preproduction samples of the generator to be furnished under this specification shall be produced prior to the manufacture of these items in production quantity. The manufacturer shall subject these generators to the preproduction inspection specified herein to determine conformance to the quality assurance provisions of this specification (see 4.2). The preproduction samples shall be fully representative of the generators to be supplied from production facilities and tooling.

3.1.1 First production sample. One of the first production generators produced on the production tooling shall be subjected to the test procedure specified in 4.3.

3.1.2 Preproduction repair parts. Individual repair parts furnished under this specification shall have met the preproduction inspection requirements of this specification after installation in a generator assembly which has then passed the preproduction inspection specified herein. Parts contained within a generator assembly that has passed the preproduction inspection shall automatically be considered as having met the preproduction inspection requirements of this specification.

3.2 Design and construction. The generator shall conform to drawing 10898795 (see 6.2).

3.2.1 Interchangeability. To provide for interchangeability, generators fabricated in accordance with this specification shall have external dimensions that fall within the envelope outline of drawing 10898795. Electrical connectors, shaft and mounting dimensions and location shall also conform to the detail requirements of the drawing.

3.2.2 Materials. Materials shall be as specified herein, on applicable standards or drawings, and in applicable specifications.

3.2.2.1 Dissimilar metals. Dissimilar metals as defined by MIL-STD-889 shall not be used in contact with each other. Where such contact is unavoidable the metal shall be protected against electrolytic corrosion. When protection is used, it shall be of such type that a low impedance path is offered to radio frequency currents.

3.2.3 Standard parts. Military standard parts shall be used whenever possible.

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3.2.4 Threaded surfaces. All screw threads shall conform to Handbook H28. The thread form, class and the number of threads-per-inch shall be as specified on the applicable standards or drawings.

3.2.5 Bearings. Bearings included in generator assemblies or supplied as repair parts shall meet the requirements of FF-B-171 together with any additional requirements specified on applicable standards or drawings. Bearing lubrication shall conform to MIL-G-23827.

3.2.6 Capacitors. Capacitors included in generator assemblies or supplied as repair parts shall meet the requirements of MIL-C-11693.

3.2.7 Brushes. Brushes included in generator assemblies or supplied as spare parts shall conform to drawing 10912456.

3.2.8 Polarity. The generator shall be designed to operate with a negatively-grounded electrical circuit.

3.3 Performance.

3.3.1 Output. When tested as specified in 4.6.2, the generator shall meet the performance requirements specified on drawing 10898795 and MS51004.

3.3.1.1 Output, performance checkpoint (heat stabilized) (see 4.6.1.5). When tested as specified in 4.6.2.1, the generator shall subsequently meet the requirements at the one stabilized performance checkpoint indicated on drawing 10898795.

3.3.2 Endurance. When tested as specified in 4.6.3, the generator shall deliner output sequentially, as shown, for 1000 hours. During this time, the generator shall require no servicing or replacement of parts. The generator shall subsequently meet the requirements of 3.3.1.1.

3.3.3 Overspeed. When tested as specified in 4.6.4, the generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings, raised commutator bars, or other damage. The generator shall subsequently meet the requirements of 3.3.1.1.

3.3.4 High temperature. When tested as specified in 4.6.5, the generator frame temperature shall nc exceed 320°F. and the generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings or raised commutator bars or other damage. The generator shall subsequently meet the requirements of 3.3.1.1.

3.3.5 Low temperature. When tested as specified in 4.6.6, the generator shall evidence no thrown solder, thrown impregnating compound, damaged insulation, raised armature windings, or raised comutator bars or other damage.

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The generator shall subsequently meet the requirements of 3.3.1.1.

3.3.6 Interpole voltage. When the generator is tested as specified in 4.6.7, the voltage measured between the paralleling terminal (D) and the negative terminal (E) shall conform to the curve shown on drawing 10898795.

3.3.7 Compounding. When the generator is tested as specified in 4.6.8, field current shall always increase when load is increased.

3 3.8 Insulation resistance. When tested in accordance with 4.6.9, the generator shall have a resistance of not less than 25 megohms.

3.3.9 Ripple. When tested in accordance with 4.6.10, the generator shall meet the ripple requirements of MIL-STD-1275.

3.3.10 Dielectric strength. When the generator is tested as specified in 4.6.11, the insulation shall evidence no loosening, cracking, charring, burning, smoking, or reduction in dielectric strength.

3.3.11 Shock resistance. After testing as specified in 4.6.12, the generator shall meet the requirements of 3.3.1.1.

3.3.12 Vibration resistance. After testing as specified in 4.6.13, the generator shall meet the requirements of 3.3.1.1.

3.3.13 Fungus resistance. After having been subjected to the test specified in 4.6.14, the generator shall meet the requirements of 3.3.1.1.

3.3.14 Corrosion resistance. After having been subjected to the test specified in 4.6.15, the generator shall meet the requirements of 3.3.1.1.

3.3.15 Sand and dust. After testing as specified in 4.6.16, the generator shall meet the requirements of 3.3.1.1.

3.3.16 Generator drive.

3.3.16.1 100-hour test. When tested as specified in 4.6.17.1, the flexible drive shall limit the armature amplitude to within plus or minus five degrees.

3.3.16.2 15-minute test. When tested as specified in 4.6.17.2, the flexible drive shall limit the armature amplitude to within plus or minus seven degrees.

3.3.17 Electromagnetic compatibility. When tested as specified in 4.6.18, the generator shall be capable of meeting the electromagnetic compatibility requirements of MIL-STD-461 for tactical vehicle components.

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3.4 Painting. All exterior surfaces of the generator assembly shall be painted, except the electrical connections, terminal box, and the extended shaft. No paint shall be applied to mating surfaces on the frame or on the ends of the generator.

3.4.1 Ferrous surfaces. Ferrous metal surfaces shall be cleaned in accordance with applicable method specified in TT-C-490. The cleaned surfaces shall be treated in accordance with type I or type III of TT-C-490. Exterior surfaces shall be given one coat of primer conforming to TT-P-636, 0.75 to 1.25 mil thick or MIL-P-8585, 0.40 to 0.60 mil thick. The finish coat shall be black gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil thick.

3.4.2 Aluminum alloy surfaces. Aluminum alloy surfaces shall be cleaned and treated in accordance with MIL-C-5541 or anodized in accordance with MIL-A-8625.

3.4.3 Cadmium plated surfaces. Cadmium plated surfaces shall conform to type II, class 3 of QQ-P-416.

3.4.4 Zinc plated surfaces. Zinc plated surfaces shall conform to type I, class 3 of QQ-Z-325.

3.4.5 Stainless steel parts and surfaces. Stainless steel parts and surfaces conforming to military specifications need not be plated or painted.

3.5 Identification of product.

3.5.1 Marking. Repair parts shall be marked in accordance with MIL-STD-130. Generator assemblies shall be marked in accordance with 3.5.2 and 3.5.3.

3.5.2 Nameplate. The generator assembly shall have a nameplate affixed to the exterior of the generator frame in a location specified on the drawing (see 3.2). The nameplate shall conform to the non-dimensional requirements of MIL-P-514. All name plates shall include the following information as a minimum:

Generator, 28 Volts DC.
300 amperes.
Military part number.
Federal stock number.
Manufacturer's identification.
Manufacturer's serial number.
US

3.5.3 Permanent mark. When specified (see 6.2), the military part number required on the nameplate shall be permanently inscribed on the

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generator frame in such location as not to be covered by the nameplate.

3.6 Workmanship. The workmanship shall be such as to assure a product free of rust, burrs, and scratches. All electrical connections shall be mechanically secured before being soldered or brazed, to preclude the possibility of becoming loose due to vibration or shock. All solder cleaning agents and flux shall be noncorrosive. All traces of flux shall be chemically neutralized and removed after soldering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Contractors quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective inspection and Quality Control system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to determine quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Preproduction inspection. When required (see 3.1 and 6.3),

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preproduction sample shall consist of four generator assemblies. Preproduction inspection and testing shall be performed by the contractor under Government surveillance and shall consist of examination for defects specified in table III and performance of tests specified in table I.

4.2.1 Failure. Failure of a preproduction sample to pass any of the examinations or tests specified herein, may be cause for the Governments refusal to witness additional examinations or tests until the faults revealed have been corrected.

4.3 First production inspection. One of the first production generators produced shall be subjected to the examination for defects specified in table III and the tests specified in table II.

4.3.1 Failure. Failure of the first production sample to pass any of the examinations or tests specified herein, may be cause for the Government to refuse to accept subsequent generators until the faults revealed have been corrected.

Table I. Order of preproduction testing

Specimen	Para. No.	Test
1	4.6.11	Dielectric strength
	4.6.2	Output
	4.6.4	Overspeed
	4.6.5	High temperature
	4.6.6	Low temperature
	4.6.13	Vibration
	4.6.15	Corrosion resistance
	4.6.16	Sand and dust
2	4.6.2	Output
	4.6.18	Electromagnetic compatibility
	4.6.17	Generator drive
3	4.6.11	Dielectric strength
	4.6.2	Output
	4.6.8	Compounding
	4.6.7	Interpole voltage
	4.6.4	Overspeed
	4.6.5	High temperature
	4.6.9	Insulation resistance
	4.6.6	Low temperature
	4.6.12	Shock resistance
	4.6.14	Fungus resistance
	4.6.10	Ripple

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Table I. Order of preproduction testing (cont.)

Specimen	Para. No.	Test
4	4.6.11	Dielectric strength
	4.6.2	Output
	4.6.4	Overspeed
	4.6.5	High temperature
	4.6.6	Low temperature
	4.6.3	Endurance
	4.6.10	Ripple

Table II. Order of first production sample testing

Paragraph No.	Test
4.6.2	Output
4.6.4	Overspeed
4.6.5	High temperature
4.6.6	Low temperature
4.6.13	Vibration
4.6.12	Shock resistance
4.6.9	Insulation resistance
*4.6.3	Endurance
4.6.10	Ripple
4.6.7	Interpole voltage

*Endurance test is required once for first production sample tests when submitted under the conditions of this specification or if generator is modified, materials changed or other fabricating processes revised on subsequent first production samples.

4.4 Quality conformance inspection.

4.4.1 Sampling for generator assemblies.

4.4.1.1 Lot formation. A lot shall consist of all generators from an identifiable production period, from one manufacturer submitted at one time for quality conformance inspection.

4.4.1.2 Sampling for quality conformance examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105 and subjected to the examinations specified in table III.

4.4.2 Examination inspection.

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4.4.2.1 Acceptable quality level. Each generator selected in accordance with 4.4.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective.

Classification	AQL
Major	1.0
Minor	2.5

Table III. Classification of defects

Categories	Defects	Method of Insp.
MAJOR		
101.	Completeness of assembly	Visual & Functional
102.	Dimensions affecting interchangeability	Standard equipment
103.	Dimensions affecting safety	Standard equipment
MINOR		
201.	Dimensions not affecting interchangeability	Standard equipment
202.	Improper finish	Visual
203.	Improper marking	Visual
204.	Poor workmanship	Visual

4.4.3 Testing inspection.

4.4.3.1 Quality conformance tests. All generators (100 percent) shall be subjected to tests in 4.4.3.1.1. Generator frame temperatures are not stabilized for these tests.

4.4.3.1.1 Generators. Each generator shall be tested in sequence in accordance with the following:

(a) Assure that armature rotates freely except for brush drag.
(b) High potential test at 500 volts, conductor to conductor and conductor to ground.
(c) Insulation resistance test - minimum resistance to ground shall be 25 MEGOHMS.
(d) Each generator shall be tested at room temperature $77^{\circ} \pm 15^{\circ}F$. for the following:

(1) Cut in speed at 28.5 ± 0.1 volts at 0 amps load (Maximum speed allowable 1425 RPM)
(2) Minimum speed at 100 amps at 28.5 ± 0.1 volts (Maximum speed allowable 1550 RPM)
(3) Minimum speed 300 amps at 28.5 ± 0.1 volts (Maximum speed allowable 1825 RPM)

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4.4.3.1.2 Rejection and retest. Generators that have been rejected may be reworked or have parts replaced to correct defects and be resubmitted for tests. Before resubmitting, detailed information or data concerning the rejection and the action taken to correct the defects shall be furnished to the Government inspector.

4.5 Control tests.

4.5.1 Sampling for control tests. Control test samples shall be selected at the rate of two generators for every 250 produced (or portion thereof), except that not more than four generators shall be selected in any 30 day period. Samples shall be from a production lot which has passed the quality conformance examination specified in 4.4.2 and quality conformance tests specified in 4.4.3.

4.5.2 Procedure. Control test samples shall be identified as to production period and subjected to the tests specified in table IV.

Table IV. Control tests in sequence

Paragraph	Test
4.6.9	Insulation resistance
4.6.2	Output
4.6.4	Overspeed

4.5.3 Failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.6 Test procedures.

4.6.1 General.

4.6.1.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. The ambient room temperature shall be maintained at $77^{\circ} \pm 15^{\circ}F$. The specimens shall be stabilized for one hour before being subjected to the tests. The generator shall be tested in combination with a blower conforming to drawing 10898759, using adapter conforming to drawing 10884044 assembled in accordance with drawing 10889713 (see 6.4.3). The test circuit of figure 1 or Government approved equivalent shall be used wherever applicable. Readings shall be corrected for variations in the ambient temperature by use of the following formula:

$$\text{Output (corrected)} = \frac{\text{Output (observed)}}{1 + .00214\,(77-T)}$$

Where T = ambient temperature °F.

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4.6.1.2 Apparatus. In addition to standard measuring equipment (see table VI, figure 1 and applicable test specifications), test apparatus shall include chambers for conducting low temperature (-80°F), high temperature (± 200°F), sand and dust tests, and a submersion tank.

4.6.1.3 **Calibration of test equipment. Test equipment accuracy shall be such as to permit measurement of 10 percent of product or test specification tolerance.** Records of calibration of test equipment at intervals sufficient to establish required accuracy shall be made available to the Government. The inspector may refuse to conduct an inspection, where accuracy of test equipment has not been established to the satisfaction of the Government. Standard measuring instrument tolerance accuracies are listed in tables VI and VII.

Table VI. Measuring instrument tolerances for qualification

Measuring instrument	Accuracy
Voltmeter DC	± 0.25 percent (30 volts-full scale)
Ammeter DC	± 0.50 percent (500 amps-full scale)
Tachometer	± 1 percent
Thermocouple system	± 2 percent (400°F. full scale)

Table VII. Calibration of quality conformance and control test instrumentation.

Measuring instrument	Accuracy
Voltmeter DC	± 0.15 Volts
Ammeter DC	± 2.50 Amps
Tachometer	± 50 RPM at 1,800 RPM
	± 200 RPM at 10,000 RPM
Thermocouple system	± 2 percent (400°F. full scale)

4.6.1.4 Frame temperature. Frame temperature shall be measured by means of a thermocouple inserted in a hole drilled in or adjacent to the pole screw nearest the cooling air outlet of the generator. The poleshoe screw shall not be loosened.

4.6.1.5 Stabilized temperature. The generator system shall be considered to be temperature stabilized when operation under constant conditions fails to cause a change in frame temperature in excess of 2°F in any five minute period of any 20 minute test.

4.6.1.6 Polarity. To determine conformance to 3.2.8, observation shall be made of the multi-range ammeter throughout the test procedure.

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4.6.2 Output. When output tests are performed the generator shall be stabilized at the highest attainable frame temperature, not exceeding 165°F above ambient temperature at 28.5 volts. The generator shall be stabilized at this temperature, unless some factor other than temperature, such as poor commutation limits the performance at the particular point. To determine conformance to 3.3.1, at least six points shall be established on the performance curve, including the following, as applicable:

(a) Cut-in speed (see 6.5.1). This speed shall be determined with no resistance in series with the field.
(b) Minimum speed at rated output (see 6.5.2). This speed shall be determined with no resistance in series with the field.
(c) Maximum operating speed (see 6.5.5). This shall be the maximum operating speed as shown on the drawing. It shall be established by adjusting the field resistance.
(d) Performance check point (see 6.5.6).

4.6.2.1 Output, performance checkpoint (heat stabilized). To determine conformance to 3.3.1.1, the generator shall be subjected to the same test procedure specified in 4.6.2 except that the output shall be determined at only the one performance checkpoint indicated on the performance curve on drawing 10898795.

4.6.3 Endurance. To determine conformance to 3.3.2, the generator shall be operated as detailed below:

TEST STEP	GENERATOR RPM (± 25 rpm)	GENERATOR VOLTS (± 0.2 volts)	GENERATOR OUTPUT AMPS (MIN)	TIME HOURS
1	2500	28.5	100	8
2	5500	28.5	200	8
3	8000	28.5	300	8
4	2500	28.5	200	8
5	5500	28.5	300	8
6	8000	28.5	100	8
7	2500	28.5	300	8
8	5500	28.5	100	8
9	8000	28.5	200	8

This cycle is repeated thru four (4) cycles at which time the generator shall then be operated with the armature shaft at an angle of 45 ± 1 degree above the horizontal for two (2) cycles. Return to the horizontal for four (4) cycles, then operate the generator with the armature shaft at an angle of 45 ± 1 degree below the horizontal for two (2) cycles. Return to the horizontal for the last two (2) cycles to complete 1000 hours. Output current and voltage, and generator speed and temperature shall be measured during the test. Subsequent to the test the generator shall be subjected to the test specified in 4.6.2.1.

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4.6.4 Overspeed. The generator shall be stabilized at maximum operating speed. The field circuit shall then be opened and the generator speed increased to the overspeed specified on the applicable drawing for one minute. The generator shall then be inspected for damage to determine conformance to 3.3.3. Subsequent to the test the generator shall be subjected to the test specified in 4.6.2.1.

4.6.5 High temperature. The generator shall be stored for 24 hours in an ambient temperature of 160° ± 5°F. The generator shall then be stabilized at average speed at rated output at ambient temperature of 160° ± 5°F. The generator frame temperature shall be recorded. After stabilization the generator shall be operated under these conditions for four hours. The generator shall then be inspected to determine conformance to 3.3.4 Subsequent to the test the generator shall be subjected to the test specified in 4.6.2.1.

4.6.6 Low temperature. The generator shall be stored for 24 hours in an ambient temperature of minus 65° ± 5°F. The generator shall then be stabilized at average speed at rated output at an ambient temperature of minus 65° ± 5°F. After stabilization, the generator shall be operated under these conditions for four hours. The generator shall then be examined to determine conformance to 3.3.5. Subsequent to the test the generator shall be subjected to the test specified in 4.6.2.1.

4.6.7 Interpole voltage. Generators shall be stabilized at the average speed at rated output. The voltage between the paralleling terminal and ground shall then be measured to determine conformance to 3.3.6.

4.6.8 Compounding. Generators shall be operated at 6,500 rpm and the output voltage adjusted and maintained constant at 28.5 ± 0.1 volts while zero, half and full-rated loads are successively placed on the generator. Field current shall be measured to determine conformance to 3.3.7.

4.6.9 Insulation resistance. To determine conformance to 3.3.8 the generator shall be subjected to the condition B, method 302 insulation resistance test specified in MIL-STD-202. During the test the generator and motor commutator brushes shall be raised off the commutator.

4.6.10 Ripple. To determine conformance to 3.3.9 the generator shall be operated in the test circuit with the battery disconnected. Volt ripple shall be observed and recorded (see 6.4.2) under the conditions listed in table VIII.

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Table VIII

Generator RPM	Generator Volts ± 0.2 Volts	Generator Output Amperes
2400	28	0
2400	28	100
2400	28	400
4000	28	0
4000	28	100
4000	28	400
6000	28	0
6000	28	100
6000	28	400
8000	28	0
8000	28	100
8000	28	375

4.6.11 Dielectric strength. To determine conformance to 3.3.10, the generator shall be subjected to a ground test of 500 volts root-mean-square (rms) at a frequency of 60 cycles-per-second (cps), applied for one minute. The potential shall be applied to the armature and field terminals with the ground brush raised off the commutator, and with the field ground terminal disconnected.

4.6.12 Shock resistance. To determine conformance to 3.3.11, the generator shall be subjected to condition A, method 205 medium impact shock test of MIL-STD-202. During the test, the generator shall be mounted on a cradle conforming to drawing 10882774. Subsequently the generator shall be subjected to the test specified in 4.6.2.1.

4.6.13 Vibration. To determine conformance to 3.3.12, the generator shall be subjected to the method 201 vibration test specified in MIL-STD-202, except that the test shall be repeated twelve times for a total duration of 72 hours. During the test, the generator shall be operated at a speed of 4,500 ± 500 rpm. During the test, the generator shall be mounted on a cradle conforming to Army drawing 10882774. Subsequently the generator shall be subjected to the test specified in 4.6.2.1.

4.6.14 Fungus resistance. To determine conformance to 3.3.13, the generator shall be tested in accordance with class 2, method B of MIL-F-13927 except that the test period shall be a continuous 90 days. Upon removal from the cabinet, examination shall be accomplished by means of removable plates or plugs, without disassembly and without disturbing any fungus growths that have developed. Upon completion of the examination and replacement of access port covers, the generator shall be operated for one hour, without load at

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6500 ± 500 rpm before being subjected to the testing specified in 4.6.2.1.

4.6.15 Corrosion resistance. To determine conformance to 3.3.14, the generator shall be tested for resistance to corrosion in accordance with test condition A of method 101 of MIL-STD-202, except that the length of the test shall be 200 hours. Prior to the performance tests, the generator shall be operated for one hour at 6,500 ± 500 rpm without load.

4.6.16 Sand and dust. To determine conformance to 3.3.15, the generator shall be subjected to test condition A of method 110 of MIL-STD-202 except that air speed shall be 2500 ± 500 feet per minute.

4.6.17 Generator drive. To determine conformance to 3.3.16, the generator shall be tested as specified in 4.6.17.1 and 4.6.17.2. The test shall be conducted on a universal-joint torsional-vibration machine which has a fly-wheel with at least 20 times the moment of inertia of the armature being tested.

4.6.17.1 100-hour test. Armature shall be subjected to a 100 hour torsional vibration with plus or minus 1 degree torsional amplitude input to the drive shaft at critical frequencies.

4.6.17.2 15-minute test. Armature shall be subjected to a 15 minute torsional vibration with plus or minus 2 degrees torsional amplitude input to the drive shaft at critical frequencies.

4.6.18 Electromagnetic compatibility. To determine conformance to 3.3.17, the generator shall be tested in conformance with the applicable provisions of MIL-STD-461.

4.7 Inspection of preparation for delivery.

4.7.1 Materials and processes. The Government inspector shall at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to requirements of Section 5 and specifications referenced herein. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The generator covered by this specification is used

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in combination with a mechanically driven blower on the M48 and M551 Tanks (see drawing 8717421). The generator in combination with electrically driven blower, Part No. 10898759 is used on the M60A1 tank (see drawing 10889713), the M88 vehicle (see drawing 10889998) and the M107 vehicle (see drawing 10914831). Differences in the various installation drawings result from varying ducting configurations. The M60A1 installation is used in the specification as the test arrangement for evaluating the basic generator. When applicable the installation assemblies are required to conform to a pressure leakage test.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction samples when required and sample quantity if different (see 3.1).
(c) Applicable drawing (see 3.2).
(d) If permanent markings are required (see 3.5.3).
(e) Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

6.3 Preproduction inspection. Preproduction inspection is required of those manufacturers who have not supplied generators conforming to the preproduction requirements of this specification during the past five years (see 4.2).

6.3.1 First production inspection. These tests are required on the first samples produced on any production contract to determine if the generator produced meets the requirements of this specification. (Applies only to those contractors who have complied with 6.3 within the last five years).

6.4 Special equipment.

6.4.1 CEC oscillographic tape has furnished satisfactory results for measuring transient characteristics.

6.4.2 Tektronix Scope, Type 545 has proven adequate for measuring voltage ripple.

6.4.3 When required the Government will furnish blower Part No. 10898759 and adapter, Part No. 10884044.

6.5 Definitions.

6.5.1 Cut-in-speed. Cut-in-speed is the lowest speed that the generator, with frame temperature stabilized will generate 28.5 volts with no external load.

6.5.2 Minimum speed at rated output. Minimum speed at rated output is the lowest speed at which rated output can be obtained at 28.5 volts.

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6.5.3 Maximum operating speed at rated output. Maximum operating speed at rated output is the highest speed at which rated output is generated at 28.5 volts.

6.5.4 Average speed at rated output. Average speed at rated output is the speed half way between the "minimum speed at rated output" and "the maximum speed at rated output" at 28.5 volts.

6.5.5 Maximum operating speed. Maximum operating speed shall be the maximum operating speed as indicated on the drawing.

6.5.6 Performance check point. The performance check point is that point on the speed output curve used as a reference point to check generator output. It shall be indicated on the drawing.

6.6 This limited coordination specification is issued for use by the Department of the Army for procurement of 300 ampere generators pending revision of MIL-G-12604 to either delete coverage of generator of this size or to provide update information relating thereto. All invitation for bids for these generators for vehicles or equipments on which they are used should refer to this specification rather than to MIL-G-12604.

Custodian:
Army - AT

Preparing Activity
Army - AT

Project No. 2920-A329

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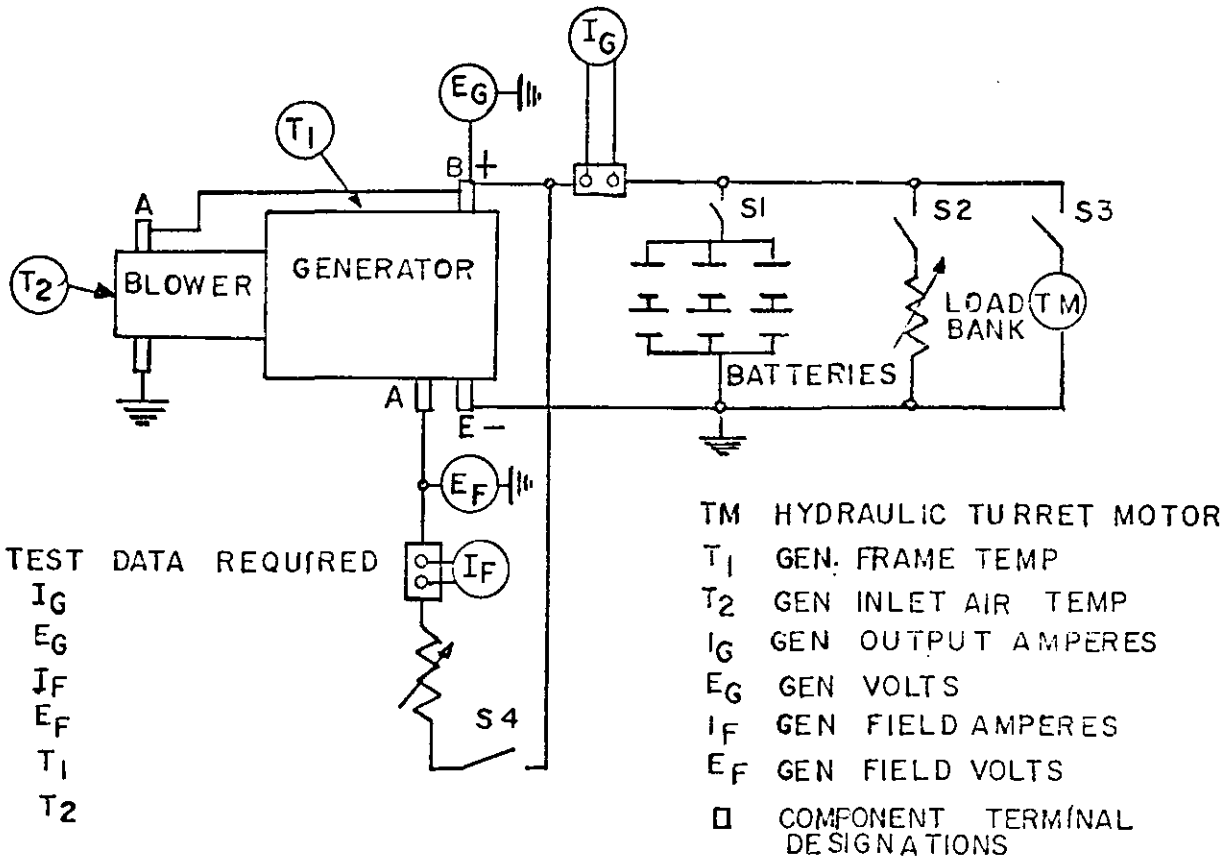


FIG. 1 TEST CIRCUIT

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-G-62061A(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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